COMP 5314-P01
Advanced Database Management Systems

Fall 2026

Group Project (20 pts)
Database Design & Implementation

Instructor: Dr. Mary Kim

September 8, 2026

MAJOR GOALS OF THE PROJECT

- Students can design and implement close to real-world Database Systems.
- Students can acquire skill sets needed for successful career in computer/information technology area such as programming MySQL, php, and web programming.
- Students can strengthen their skills in communication, teamwork, and documentation.

MAJOR STEPS FOR THE PROJECT

1. Make three groups (3 or 4 members each group). Choose a database system your group wants to develop. Group table is prepared on Canvas module where you can write group name, member names, group leader and project topic. This table will be finalized at September 22, 2026, 5 pm.
2. Prepare mid-term presentation on October 6, 2026.
   - Create a mission statement for your database system, i.e., purpose of the system -- improvement of existing services or development of new services.
   - Create an ER diagram (conceptual design) and a Table diagram (logical design) for your DB system. Your database should have at least 7 tables.
   - Identify five main business questions that your database should answer. Do not include SQL code.
   - Set up timeline for the project and assign roles to each member.
   - Upload a ppt file (one per group) into Canvas by October 6, 5:00 pm.
3. Prepare final presentation and final paper: presentation and paper due November 17, 2026 .

- Implementation
  - ✓ Implement using phpMyAdmin and insert sufficient records for testing.
  - ✓ Implement SQL queries to answer at least **7 complex business questions**. All these queries need to involve at least **2 tables** (which means combinational queries)
  - ✓ Create tables including **Check** constraint (at least 3 tables), **PK, FK** constraint with proper data type and size definition.
  - ✓ Using phpMyAdmin, **implement three triggers, three procedures or functions, and three cursors.**
  - ✓ Implement a web interface using php to connect with the database. Display the results of SQL queries in proper web formats.
- Presentation & Paper
  - ✓ Prepare slides for group presentation (20 min).
  - ✓ Final paper should include introduction, overview of the system, table diagram, implementation methods, main queries examples, screen capture of executions (major trigger/procedure/cursor examples), discussion & conclusion, what you learned through this project, and references.
  - ✓ Upload a zip file: paper in pdf + source code (sql, php) + readme.txt (including authors, project overview, file structure, etc.)
  - ✓ Upload a zip file into Canvas by November 17, 2026, 5:00 pm

MAIN RESULTS OF PROJECT

- Two Presentations

Mid-term	October 6, 2026	15 min
Final	November 19, 2026	20 min

- Final paper November 19, 2026, 5:00 pm
  12-13 pages (not including cover and contents of table), Times New Roman, 12 pts, 1.15 lines spaces

PROJECT GRADING POLICY (20 POINTS)

- Mid-term presentation – 30%
- Final presentation – 30%
- Final paper – 30%
- Peer Assessment – 10%

- ✓ Presentation evaluation consists of the instructor (60%) and audience (40%)
- ✓ Group members receive the same grade except peer assessment
- ✓ Audience will evaluate the other groups' presentation

* Best project will get 1 extra credit added to the final grade to all members.

Make a group of 3-4 members. Choose a database system you want to develop. Fill out group project table below by September 22, 2026, 5 pm. Any left-out student by then will be assigned to a random group.

Project Group Table

	Group Name	Group Members	Group Leader	Topic
1				
2				
3				